Road Accident Data Analysis, Visualization and Severity Prediction

EDA Assignment 2 - Report

Team 6

Dataset: synthetic road accident records, 8 Tamil Nadu cities, 2023-2024

Records (original)	Records (cleaned)	Fatal accidents	Best model
1,560	1,500	200 (13.3%)	Gradient Boosting

Files accompanying this report: EDA_Assignment_2_Team6.ipynb (code and outputs), Team 6 - original data ,cleaned data.xlsx (raw and cleaned data), model_comparison.csv, feature_importance.csv, EDA_Assignment_2_Presentation.pptx, README.txt.

1. Objective

Road accidents cause serious injury and loss of life. The aim of this assignment is to (a) clean a messy accident dataset, (b) explore when, where and why accidents happen and which conditions make them more severe, and (c) build and compare machine-learning models that predict accident severity (Minor, Serious or Fatal) from the conditions at the scene.

Note: the data is randomly generated for practice. All findings describe this synthetic dataset and are illustrative only.

2. Dataset description

Each row is one accident. The original file has 1,560 rows and 15 columns.

Column	Meaning
accident_id, date, time	Unique id and when the accident happened
city, road_type	Eight cities; National Highway, State Highway, City Road, Rural Road
weather, light_condition	Clear / Rain / Fog / Cloudy; Daylight / Dusk-Dawn / Dark
vehicle_type, cause	Main vehicle involved and the primary cause
severity (target)	Minor, Serious or Fatal
num_vehicles, num_casualties	Vehicles involved; people injured or killed
driver_age, driver_gender, speed_limit	Driver details and the posted speed limit

3. Data cleaning

The original data was deliberately messy, as real data usually is. The table lists each problem and the fix applied. After cleaning there are 1,500 rows and no missing values.

Problem in original data	Action taken
60 duplicate rows	Dropped (1,560 to 1,500 rows)
Mixed case and extra spaces in text columns (e.g. 'CHENNAI', ' Madurai ')	Trimmed and standardised to Title Case
'Trichy' and 'Tiruchirappalli' for the same city; 'M'/'F' vs 'Male'/'Female'	Mapped to one label each
Three different date formats	Parsed into a single datetime column
Impossible ages (-5, 0, 150, 210) and 'unknown' text	Set to missing, filled with the median age
Impossible casualty counts (-1, 45, 99)	Set to missing, filled with the median for that severity
Missing time (49), weather (63), gender (49), city (16); cause 'Unknown' (46)	Time inferred from light condition; others filled with 'Unknown'
Missing speed limit (66)	Filled with the median for the same road type
No time-based fields	Derived hour, weekday, month, is_weekend, age_group, time_of_day

4. Exploratory data analysis

4.1 Severity

Of 1,500 accidents, 793 (52.9%) were minor, 507 (33.8%) serious and 200 (13.3%) fatal. The accidents caused 2,333 casualties in total. The classes are imbalanced, which matters for modelling.

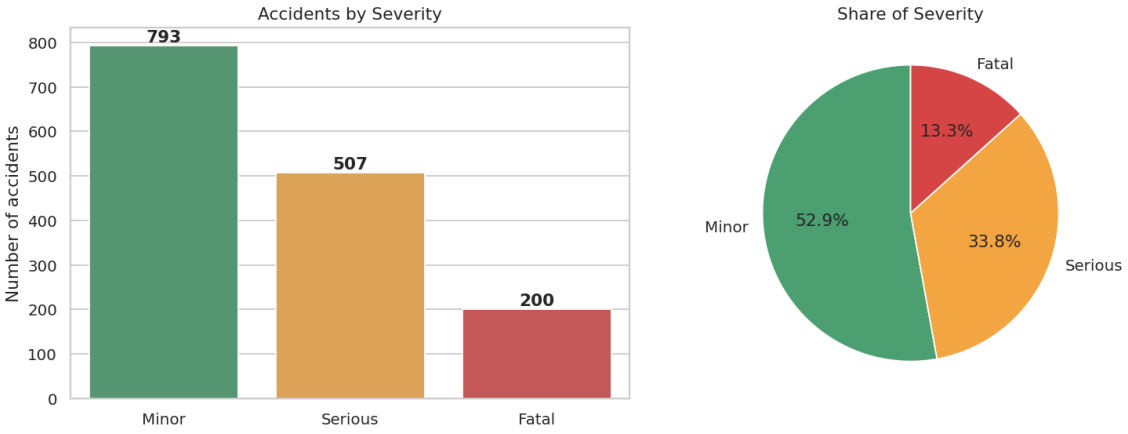


Figure 1. Accidents by severity.

4.2 When do accidents happen?

Accidents peak in the evening: the four hours from 17:00 to 20:59 hold 28% of all accidents, with the single busiest hour at 19:00. A second, smaller peak occurs in the morning commute (about 07:00-08:00).

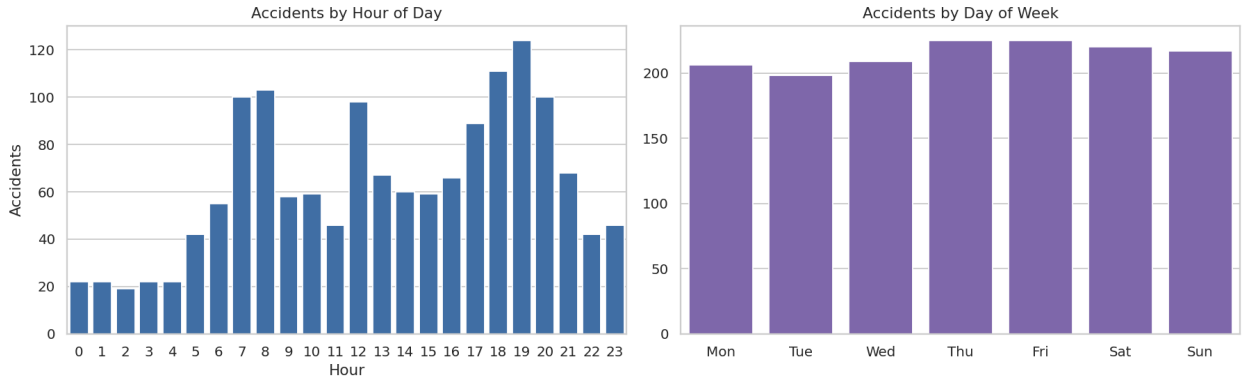


Figure 2. Accidents by hour of day and by weekday.

4.3 Where do accidents happen?

Chennai records the most accidents (434, 29% of the total), followed by Coimbatore and Madurai. However, the *share of accidents that are fatal* is highest in Tiruchirappalli (16.2%) and Coimbatore (15.9%) and lowest in Tirunelveli (9.5%). Volume and severity therefore point to different cities.

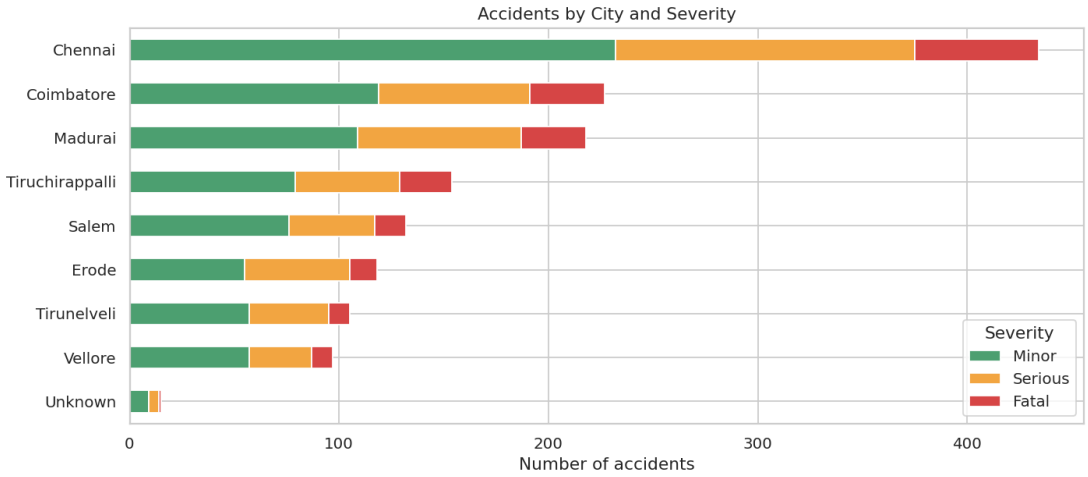


Figure 3. Accidents by city and severity.

4.4 Causes and vehicles

Over-speeding is the most frequent cause. Two-wheelers are involved in 42% of accidents (633), far more than cars. Frequency and danger differ by cause: 25.9% of drunk-driving accidents and 22.4% of over-speeding accidents were fatal, compared with about 6% for distracted driving, signal violations and wrong-side driving.

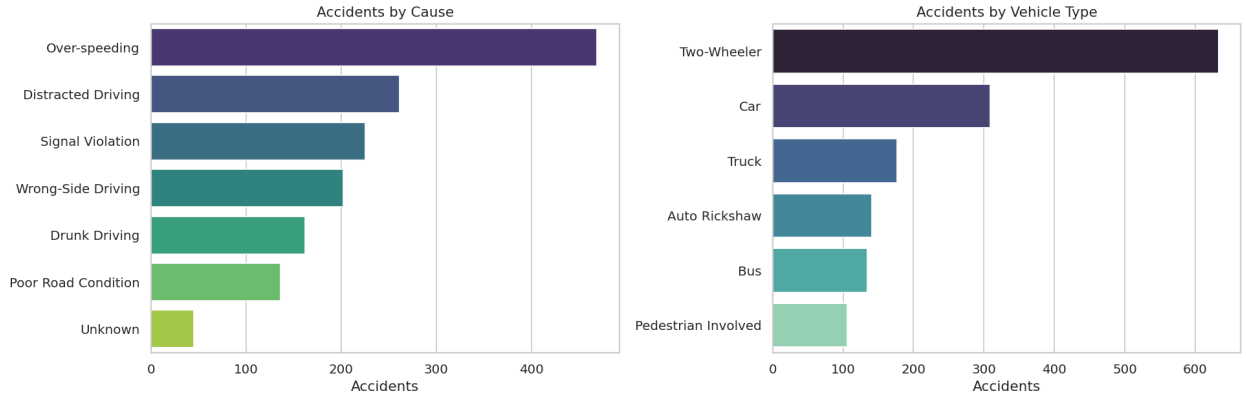


Figure 4. Accidents by cause and by vehicle type.

4.5 Weather, light and road type

Conditions that reduce visibility or grip raise the fatal share sharply. Fog (26.3%) and rain (20.3%) are far above clear weather (9.9%). In the dark 23.3% of accidents are fatal versus 7.8% in daylight, about three times higher. National Highways are the most dangerous road type (24.4% fatal) compared with city roads (6.0%) and rural roads (4.5%).

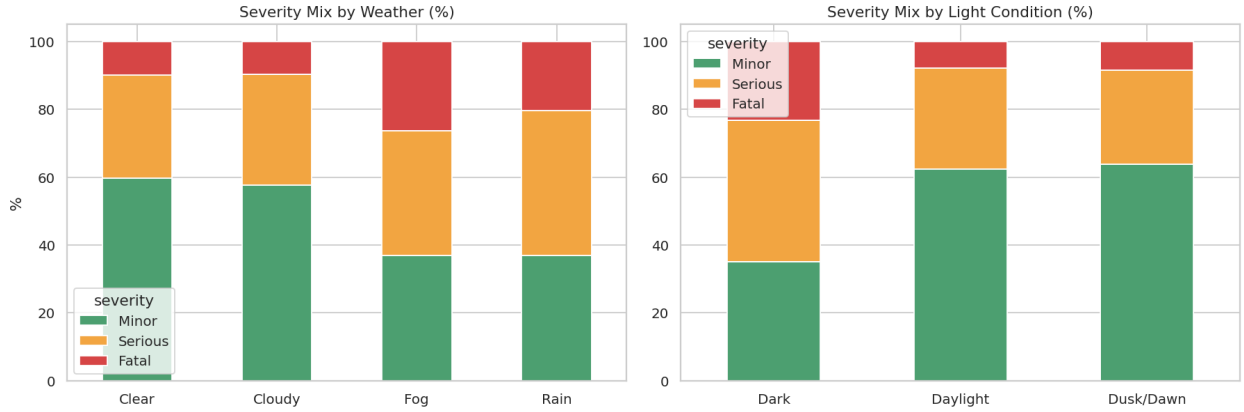


Figure 5. Severity mix by weather and by light condition (percent of accidents).

5. Predictive modelling

Task. Predict severity (Minor / Serious / Fatal) from city, road type, weather, light, vehicle, cause, driver age and gender, speed limit, number of vehicles, hour, month and weekend flag. **num_casualties was left out** because it is an outcome of the accident, and using it would be data leakage.

Method. One-hot encoding for categories and standard scaling for numbers inside a scikit-learn Pipeline; 75/25 stratified train-test split; 5-fold cross-validation on the training set. Five models were compared. Because severity classes are imbalanced, macro-F1 (the average F1 over the three classes) is the main metric.

5.1 Model comparison

Model	CV accuracy	Test accuracy	Weighted F1	Macro F1
Gradient Boosting	0.588	0.637	0.629	0.566
Logistic Regression	0.572	0.605	0.614	0.560
Random Forest	0.574	0.592	0.594	0.531
Decision Tree	0.501	0.531	0.532	0.458
KNN (k=15)	0.588	0.571	0.530	0.426

Gradient Boosting performs best (test accuracy 63.7%, macro-F1 0.57); Logistic Regression is close behind. Always predicting 'Minor' would give about 53% accuracy, so the best model is better than the naive baseline, but only moderately. Minor accidents are predicted well (F1 0.76), Serious ones less so (0.51), and Fatal ones least (0.44) because they are the rarest class.

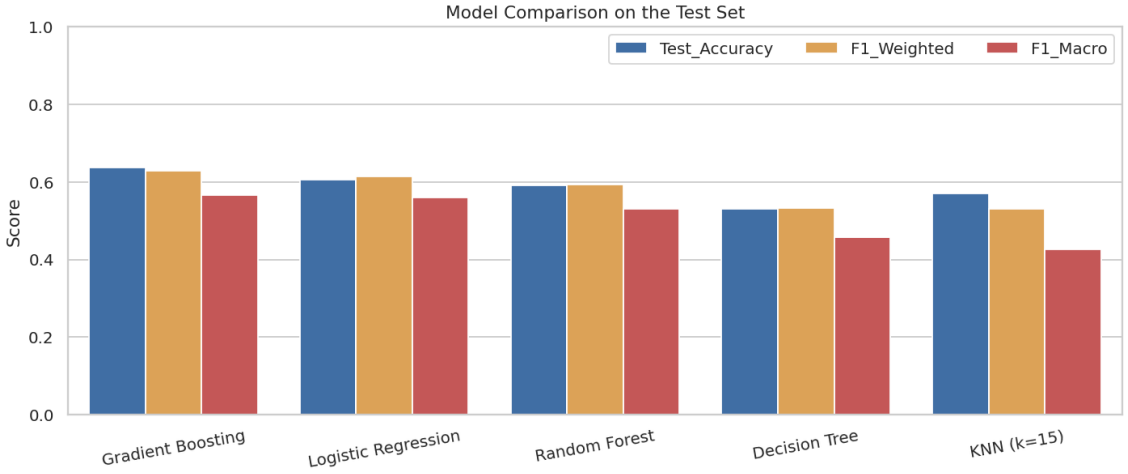


Figure 6. Model comparison on the test set.

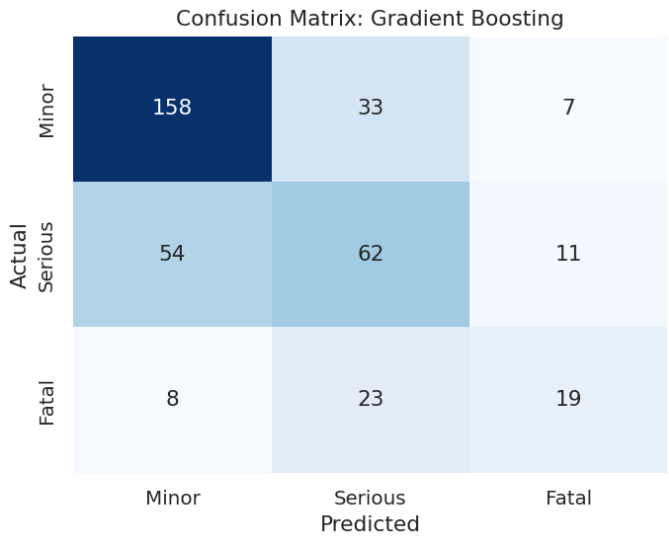


Figure 7. Confusion matrix of the best model.

5.2 Which features matter most?

Permutation importance measures how much the macro-F1 falls when one feature is randomly shuffled on the test set. Cause is the strongest predictor, followed by vehicle type, light condition and road type. This agrees with the exploratory analysis. Smaller values (for example driver age) have wide error bars (see Importance_Std in feature_importance.csv) and should not be over-interpreted.

Rank	Feature	Importance (drop in macro-F1)
1	cause	0.1008
2	vehicle_type	0.0552
3	light_condition	0.0515
4	road_type	0.0503
5	driver_age	0.0452
6	weather	0.0411
7	hour	0.0300
8	city	0.0267

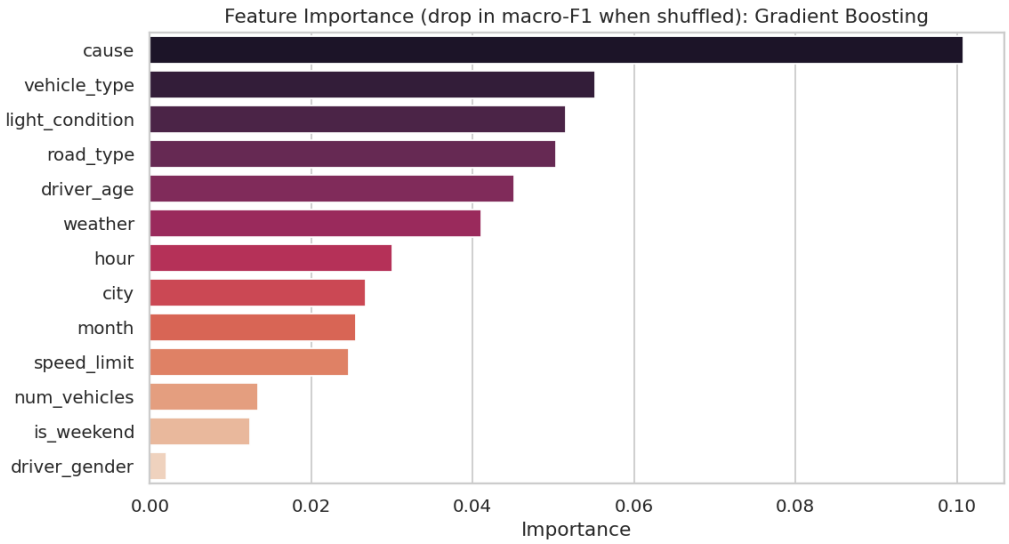


Figure 8. Permutation feature importance.

6. Conclusions and recommendations

- Evening rush hour (17:00-21:00) accounts for 28% of accidents; targeted traffic management and police presence in these hours would reach the largest number of cases.
- Severity is driven by conditions rather than volume: darkness, fog, rain and National Highways each roughly double to triple the fatal share. Better lighting, fog/rain advisories and speed control on highways are the most evidence-backed actions here.
- Drunk driving and over-speeding have the highest fatality shares (about 26% and 22%); breath-test drives and speed cameras are justified.
- Two-wheelers dominate the accident count, so helmet enforcement and awareness campaigns should focus on them.
- Chennai needs attention for accident volume, while Tiruchirappalli and Coimbatore need attention for severity.
- A Gradient Boosting model gives a useful but limited severity prediction (macro-F1 0.57); it could support risk scoring but not individual decisions.

7. Limitations

- The data is synthetic, so patterns are illustrative and do not describe real road conditions.
- Important real-world factors (impact speed, helmet or seat-belt use, vehicle condition, emergency response time) are not in the data, which limits predictive accuracy.
- Imputation with medians and 'Unknown' labels is simple; it can slightly reduce variation in the cleaned data.
- The test set is small (375 rows, only 50 fatal cases), so model scores have some uncertainty.